Variational and Optimization-based Approaches for Event-Based Vision Applications

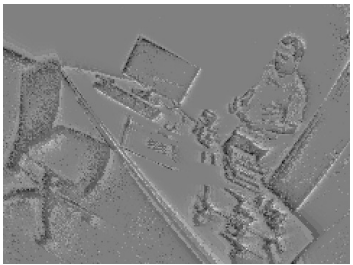
From unobserved modes to well-posed estimators

Daniel Blauvelt

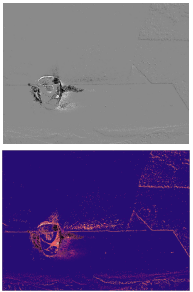
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Dissertation Defense ▪ Fall 2026

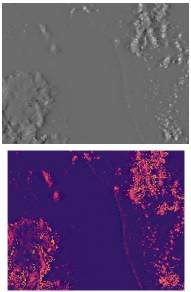
Scenes recovered from brightness changes



Office, handheld; ≈ 2.5M events



Bullet hits a cup; ≈ 350K events



ISS (Falcon Neuro); ≈ 2M events

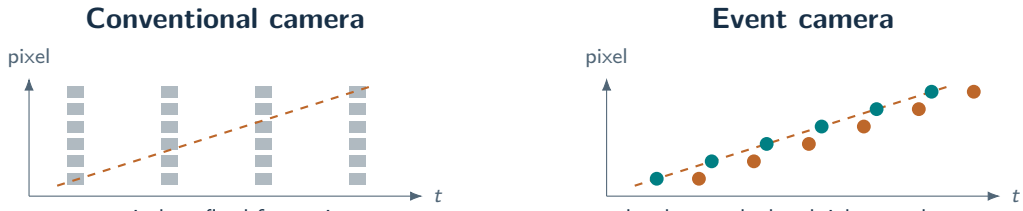
Cup and ISS: grayscale (top) and false color (bottom) of the same reconstructed frame.

Each pixel reports when its brightness changes, with microsecond timing and low power [1].

This dissertation recovers images, motion, and depth from these changes, with guarantees on the estimates and computation simple enough for on-board use.

Dissertation: Figure 2.7, Figure 2.8, and Figure 2.9; Section 2.3.3. Data: office [36], cup [37], Falcon Neuro [38].

Why an event camera? Each pixel reports its own changes

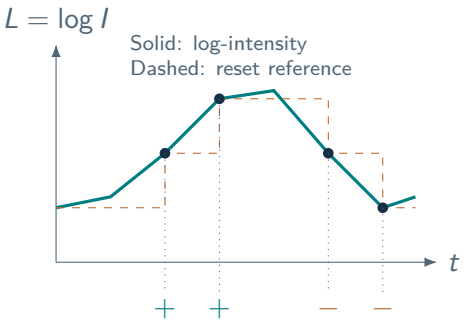


Dashed line: the leading edge of a moving bright bar. Teal: brightening (+); amber: darkening (−).

- Microsecond timestamps, about 140 dB dynamic range versus 60 dB, low power, little motion blur [1].

Dissertation: Chapter 2, neuromorphic camera overview; Gallego et al. [1]. Diagram is schematic.

An event camera: each pixel reports threshold crossings



An event is a tuple

$$e_k = (\mathbf{x}_k, s_k, p_k), \qquad p_k = \pm 1.$$

For the ideal threshold model,

$$L(\mathbf{x}_k, s_k) - L(\mathbf{x}_k, s_k^-) = p_k c_{\mathrm{th}}.$$

The preceding reset s_k^- is **local to that pixel**.

Dissertation: Eq. 2.1; Eq. 3.3; ideal threshold model [20]. Diagram is schematic.

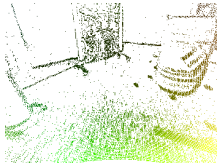
From events to images, motion, and depth



Image reconstruction

Ch. 2

Grayscale: reconstructed value.

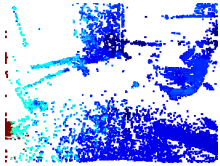


Optical-flow estimate

Ch. 3; event-active pixels



hue: direction
brightness: size
(full at ≈ 14 px)



Stereo-depth estimate

Ch. 4

depth (jet), near to far

0.5 — 8 m

MVSEC [35] indoor_flying2, drone-mounted event cameras, near 47.6 s; stereo depth uses both.

Dissertation: Methods: Chapter 2, Chapter 3, and Chapter 4; MVSEC [35].

Three estimators, each well posed

Reconstruction (Ch. 2)	One tridiagonal solve per pixel; unique and stable for every positive weight.
Optical flow (Ch. 3)	Three event representations, one quadratic theory; unique flow when edges span two directions.
Stereo depth (Ch. 4)	A unique inner solve at each candidate depth; competitive MVSEC accuracy with the most scored points.

The common question: which quantities do the events determine, and which does the model supply?

Dissertation: Table 5.1; Theorem 2.2.5, Theorem 3.6.1, Theorem 4.6.3.

Start with one pixel: integrated changes on a temporal chain

Choose a unit-mass temporal kernel ϕ_{s_k}:

$$e(t) = \sum_{k \text{ at this pixel}} p_k \phi_{s_k}(t).$$

$$y_i = \int_{t_i}^{t_{i+1}} e(t)\, dt.$$

The reconstruction times t_i are chosen by the algorithm.

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix}.$$

$$\mathbf{Af} \approx \mathbf{y}.$$

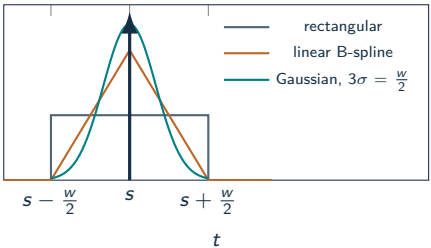
$$\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}.$$

Events determine increments, leaving a constant free.

In scene terms: each pixel's log-brightness history is rebuilt on its own from how much it rose or fell in each interval. The model chooses the reference level; a pixel without events stays at that level, whatever its brightness.

Dissertation: Eq. 2.4, Eq. 2.7, and Lemma 2.2.4.

Temporal kernels: how far each event's influence spreads



Arrow: delta kernel (pure binning). Common support width w; schematic.

Each kernel has unit mass, is a translate, and has compact support:

$$\int_{\mathbb{R}} \phi_s = 1, \qquad \phi_s(t) = \phi_0(t - s).$$

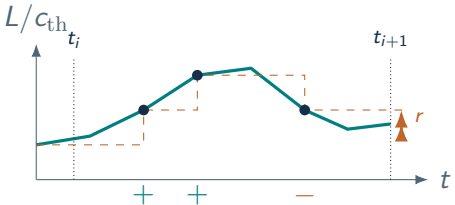
Each event splits its sign p_k across the bins it overlaps:

$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt, \quad \sum_i y_i = \sum_k p_k$$

when every support lies inside $[t_0, t_N]$.

Dissertation: Proposition 2.2.1, Proposition 2.2.2; Table 2.2.

From threshold crossings to an approximate observation



Solid: L/c_{th}. Dashed: reference at the last event.

Ideal sensor: each event marks a log-brightness change of about one threshold c_{th}.

$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt \;\approx\; \frac{L(t_{i+1}) - L(t_i)}{c_{\mathrm{th}}}.$$

Exactly, for any times $a < b$ at one pixel:

$$\frac{L(b) - L(a)}{c_{\mathrm{th}}} = y(a,b) - \underbrace{B(a,b)}_{\text{kernel spill at bin edges}} + \underbrace{\frac{1}{c_{\mathrm{th}}} \sum_{a < s_\ell \le b} \varepsilon_\ell}_{\text{overshoot}} + \underbrace{r(b) - r(a)}_{\text{sub-threshold change}}.$$

Dissertation: Physical connection (p. 24), Eq. 2.29, arbitrary-endpoint identity; threshold model [20]. Schematic diagram.

A unique, stable estimate for every positive weight

For fixed $\mathbf{\Lambda} = \mathrm{diag}(\lambda_i)$ with every $\lambda_i > 0$,

$$\mathbf{f}^* = \arg\min_{\mathbf{f}\in\mathbb{R}^n} \tfrac{1}{2}\|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \tfrac{1}{2}\|\mathbf{\Lambda}\mathbf{f}\|^2, \qquad \mathbf{K}_{\mathrm{time}} = \mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda} \succeq \lambda_{\mathrm{min}}^2\mathbf{I}.$$

- **Unique:** the normal equations $\mathbf{K}_{\mathrm{time}}\mathbf{f}^* = \mathbf{A}^\top\mathbf{y}$ have exactly one solution.
- **Stable:** $\|\delta\mathbf{f}^*\| \le (2/\lambda_{\mathrm{min}}^2)\,\|\delta\mathbf{y}\|$, so one spurious event moves a pixel's history by at most $2/\lambda_{\mathrm{min}}^2$.
- **Reference level:** $\mathbf{1}^\top\mathbf{\Lambda}^2\mathbf{f}^* = 0$; quiet intervals carry the largest weights and set the level.

Dissertation: Eq. 2.8, Theorem 2.2.5, Proposition 2.2.7; Tikhonov regularization [27].

One tridiagonal solve per pixel

$\mathbf{A}^\top\mathbf{A}$ has diagonal $(1, 2, \ldots, 2, 1)$ and off-diagonals -1, so

$$\mathbf{K}_{\text{time}} = \mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda}$$

is tridiagonal and diagonally dominant.

- Thomas algorithm: $O(N)$ operations per pixel, no pivoting.
- Pixels are independent; a pixel without events returns $\mathbf{f} = 0$.
- The newest frame comes from the forward sweep alone: constant work per pixel for each new interval.

Historical laptop measurements

Earlier penalty convention; current-estimator timings require a rerun.

processing time ÷ recording time; 100 ms windows, $N = 50$

Sensor	Pixels	Events/s	1 thr.	All
DAVIS346	90k	0.25 M	0.03	0.01
EVK4	922k	13 M	2.5	0.67

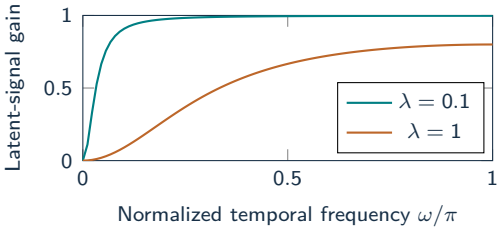
Earlier cup implementation, 500k events, 150 frames: 39.6 s → 0.22 s; agreement within 10^{-15} for that earlier estimator.

Dissertation: Eq. 2.7, Theorem 2.2.5; Thomas algorithm [32]; timings: edge implementation, October 2026.

What the penalty does to temporal information

For scalar $\mathbf{\Lambda} = \lambda \mathbf{I}$ and noiseless data $\mathbf{y} = \mathbf{A}\mathbf{f}_{\mathrm{true}}$:

$$g_k = 4 \sin^2 \frac{\pi k}{2n}, \qquad H_k = \frac{g_k}{g_k + \lambda^2}, \qquad H_0 = 0.$$



The baseline is removed.

Low temporal modes are attenuated most.

A unique estimate does not restore the missing physical baseline.

In the images: slow brightness drift fades; passing edges remain.

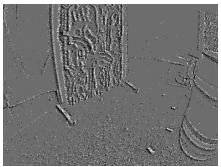
Dissertation: Lemma 4.4.1, Theorem 4.6.21; cosine eigenbasis [81]. Curves: continuous extension.

MVSEC comparison with Antil–Sayre bilevel

MVSEC [35] indoor flying 2



Reference



Event only

This chapter: events only → activity weights $\mathbf{\Lambda}(dc)$ → fixed convex solve.

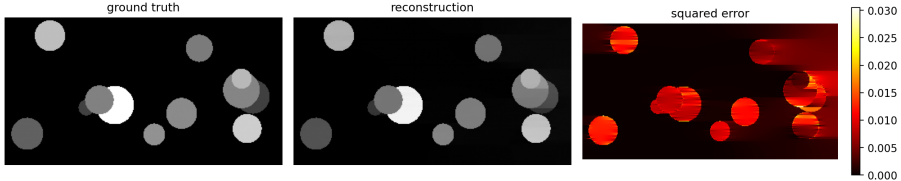
Antil–Sayre [2]: events and frames → outer optimization over the blur-kernel parameter z.

	PSNR (dB)		CN	
Sequence	Bilevel	Events	Bilevel	Events
Indoor	15.37	10.85	39755	25.51
Outdoor	8.41	8.71	39755	27.10

PSNR: similarity to a reference frame (dB). CN: condition number of each method's inner linear system.

Dissertation: Table 2.8 and Table 2.9. Antil–Sayre [2], Sec. 3, Eqs. (3.1)–(3.3). Images: MVSEC indoor flying 2.

Shared motion couples temporal profiles to one scene



Simulated events (v2e [46]); the camera window shifts 1 px per view. Left: scene. Center: shared-scene estimate (smoothed-TV penalty). Right: squared error.

Chapter 2: each pixel has its own temporal profile.

Shared scene: known motion P maps one static scene θ into every view.

$$\underbrace{B_{\mathrm{SAR}}}_{\text{scene-to-event operator}} = (A \otimes I)\, P.$$

Dissertation: Figure 2.10, Eq. 2.47, Proposition 2.4.1; joint ESAR work [45]; synthetic aperture [98].

The aperture problem: what a local view reveals

An **aperture** is a restricted observation window: a small view of an edge [66].

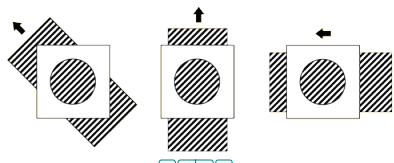
Through the windows



The visible stripe motions look the same.

Movie 5 source (GIF)

Reveal the full patterns



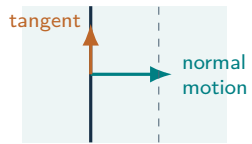
The patterns translate in different directions.

Movie 6 source (GIF)

Perceptual illustration: Wolfe et al., *Sensation and Perception*, 4th ed., Sinauer © 2015. Activity 8.2, Movies 5–6.
Dissertation: Proposition 3.3.4 and the following aperture discussion.

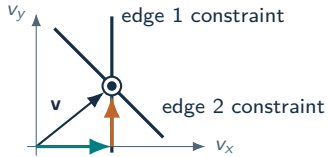
Motion introduces a new invisible direction

Image: one moving edge



A straight edge reveals its normal speed.

Velocity: what the events allow



Edge 1 fixes the normal part; edge 2, if it shares the motion, fixes the tangent part.

Flow model: events give each edge's normal speed; neighbors moving alike supply the rest.

Requirement (main theorem): edges in two orientations somewhere, so their constraint lines cross.

Intersection of constraints [Adelson & Movshon 1982]. Dissertation: Proposition 3.3.4; requirement in Theorem 3.6.1.

Three representations, one quadratic variational problem

Rename the flow data matrix $\mathbf{B}$ to distinguish it from temporal $\mathbf{A}$:

$$E_{\mathbf{B},b}(\mathbf{v}) = \frac{1}{2}\int_\Omega (\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}) + \frac{\alpha^2}{2}\int_\Omega |\nabla \mathbf{v}|^2.$$

Representation	$\mathbf{B}$	b	Edge measurement from
Time surface	$\nabla\mathcal{T}\nabla\mathcal{T}^\top$	$\nabla\mathcal{T}$	crossing times
Voxel projection	$\nabla a \nabla a^\top$	$-a_s\nabla a$	transported event activity
Structure tensor	$\widetilde{\mathbf{J}}$	$-\mathbf{j}_t$	gradients pooled over a window

In scene terms: $\mathbf{B}$ records which motion directions the edges measure, and how strongly; b carries the measured values; the α term asks neighbors to move alike.

Dissertation: Theorem 3.6.1 and its three corollaries; data-plus-smoothness form [16].

The main flow theorem: global information controls constants

Unified well-posedness theorem

Let $\Omega \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz; $\alpha > 0$. Assume

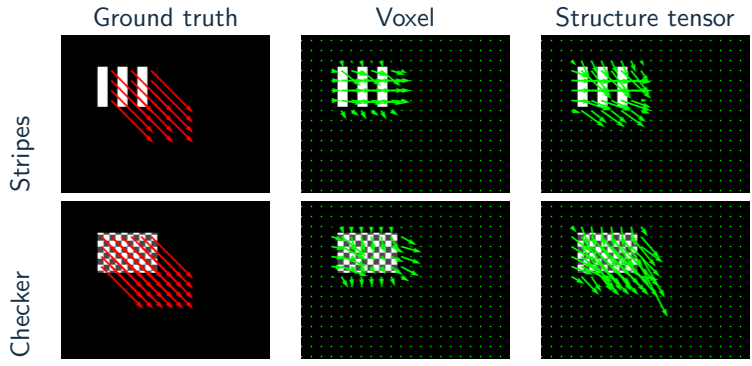
$$\mathbf{B} \in L^\infty, \quad \mathbf{B} = \mathbf{B}^\top \succeq 0 \text{ a.e.}, \quad b \in L^2, \qquad \boxed{\int_\Omega \mathbf{B}(\mathbf{x})\, d\mathbf{x} \succ 0}.$$

Then $E_{\mathbf{B},b}$ has a unique minimizer in $H^1(\Omega;\mathbb{R}^2)$.

The integrated matrix must identify **every constant velocity direction**.
In scene terms: edges in two or more orientations somewhere, so their constraint lines cross; neighbors supply the motion where $\mathbf{B} = 0$.

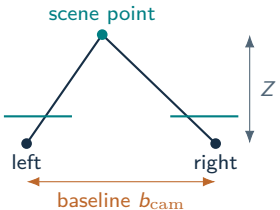
Dissertation: Theorem 3.6.1, Eq. 3.59; classical quadratic-flow theory [57].

Known-motion examples make the aperture problem visible



Dissertation: Figure 3.4; v2e [46] simulation, same window and arrow scale.

Stereo depth: competitive accuracy, most scored points



$$Z = \frac{f_{\mathrm{cam}} b_{\mathrm{cam}}}{d}, \qquad d = x_L - x_R > 0.$$

Each candidate d pairs the two views, and one shared temporal profile is fitted to both. For fixed d and frozen weights the inner solve is unique.

Method	Mean depth error (cm) ↓ FI1	FI2	FI3	FI4	log RMSE ×100 ↓	Scored points (M)
WLS (ours)	73.83	**58.35**	70.33	59.24	**44.54**	**2.66**
Piatkowska adaptation [78]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20

MVSEC indoor flying 1–4; each method scored on its own estimates in $[0.5, 8]$ m. Three more methods in the backup table.

Dissertation: Section 4.5, Eq. 4.11, Theorem 4.6.3; Table A.5; MVSEC [35]. Geometry is schematic.

Research opportunities

Sensor modeling

Measure the error terms of slide 9 on real sensors; the stability bounds turn them into estimate bounds.

Learned weights

Learn $\mathbf{\Lambda}$ in an outer problem [Antil–Di–Khatri 2020]; for $\mathbf{\Lambda} \succ 0$, $\mathbf{f}^*(\mathbf{\Lambda})$ is smooth.

Shared scenes

Find the motion-invariant modes of B_{SAR} for a trajectory (ESAR [45]); apply to orbital streams.

Edge hardware

Update only pixels with events; measure time and power on a Raspberry Pi 5 and edge sensors.

Dissertation: Chapter 5, Future Directions; physical connection (p. 24); Proposition 2.4.1. External paper cited above.

What the data constrain and what the models prove

Problem	What the observations constrain	What the mathematical model establishes
Temporal reconstruction	Represented brightness increments	Unique regularized history for each fixed positive weight matrix, with perturbation bounds
Optical flow	Event-derived directional motion constraints	Unique H^1 minimizer under integrated identifiability
Stereo depth	Paired temporal event-feature observations	Unique frozen-weight inner solve; the disparity comes from a separate comparison of candidates

Takeaway. Under the stated hypotheses, regularization makes each estimator determinate; accuracy also depends on the representation.

Dissertation: Theorem 2.2.5, Theorem 3.6.1, Theorem 4.6.3; Table 5.1.

Which directions do the observations control, and which are selected by the model?

Thank you.

Must every temporal penalty entry be positive?

For arbitrary $\boldsymbol{\Lambda}$, the exact algebraic criterion is

$$\mathbf{A}^\top \mathbf{A} + \boldsymbol{\Lambda}^\top \boldsymbol{\Lambda} \succ 0 \quad \Longleftrightarrow \quad \ker \mathbf{A} \cap \ker \boldsymbol{\Lambda} = \{0\}.$$

Why?

The quadratic form vanishes precisely when both $\mathbf{A}\mathbf{z} = 0$ and $\boldsymbol{\Lambda}\mathbf{z} = 0$.

On a connected first-difference chain, $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$.

- A single nonzero diagonal penalty entry removes that constant mode.
- Fixing one reference value or imposing zero mean also removes it.

Takeaway. Any choice of reference level resolves the mathematical ambiguity; a measured reference would add physical information.

Dissertation: Lemma 2.2.4; discussion following Lemma 4.4.1.

Kernel performance depends on the scene and adaptive weights

	MVSEC indoor		MVSEC outdoor	
Kernel	PSNR	CN	PSNR	CN
B-spline	8.86	22.16	6.72	97.92
Delta	7.70	1375.75	7.97	53.34
Gaussian	8.84	20.22	6.81	98.90
Rectangular	8.24	423.59	7.20	148.33

- The temporal operator stays the same; represented data and adaptive penalties change.
- Better conditioning is not the same as smaller reconstruction bias.
- Reference-image scores do not identify an absolute intensity baseline from events.

Dissertation: Table 2.4, Table 2.5.

What the reconstruction scores measure

Quantity	What it compares	What it establishes
PSNR / SSIM	Reconstruction and reference images	Image agreement under the scoring convention
TC / TG	Temporal changes in the output	Smoothness of the returned sequence
ERC	Magnitudes of reconstructed changes and represented events	Correlation with event activity
CN	Sensitivity of a specified matrix	A numerical conditioning diagnostic

A simple check on interpretation

A constant sequence has $\mathrm{TC} = 1$ and $\mathrm{ERC} = 0$ under the stated conventions.

Takeaway. Consider reference fidelity, event consistency, and conditioning together. Smoothness alone rewards a static output.

Dissertation: Evaluation metrics (p. 31); Table 2.3; SSIM [40].

Stable estimates can still have regularization bias

For fixed scalar $\lambda > 0$, $\mathbf{y} = \mathbf{A}\mathbf{f}^\dagger + \boldsymbol{\eta}$, and $\mathbf{K} = \mathbf{A}^\top\mathbf{A} + \lambda^2\mathbf{I}$,

$$\mathbf{f}_\lambda - \mathbf{f}^\dagger = -\lambda^2\mathbf{K}^{-1}\mathbf{f}^\dagger + \mathbf{K}^{-1}\mathbf{A}^\top\boldsymbol{\eta}.$$

The two terms are regularization bias and propagated observation noise.

Takeaway. A stable estimate need not equal the physical signal.

Dissertation: Theorem 2.2.5, Proposition 2.2.7; direct bias-noise identity.

The physical event stream loses more than one baseline

The binned increment matrix has $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$, but the ideal sensor can lose time-varying signals too.

$$L_1(t) = L_0, \qquad L_2(t) = L_0 + a\sin(2\pi t/T), \qquad 0 < a < c_{\mathrm{th}}, \quad t \in [0, T].$$

Both start at L_0, emit no events, and have positive intensities $I_j(t) = e^{L_j(t)}$.

Takeaway. The chain nullspace describes the discrete increment surrogate, not every ambiguity of the physical sensor.

Dissertation: Eq. 2.1, Lemma 2.2.4; direct ideal-sensor example.

Frozen weights give a unique, stable inner solve

Fix d, $\mathbf{W}_L, \mathbf{W}_R \succ 0$, $\mathbf{\Lambda} \succ 0$, and $\gamma \geq 0$. Define

$$\mathbf{K}_c = \mathbf{A}^\top \mathbf{W}_c \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}, \qquad \mathbf{M}_W = \begin{pmatrix} \mathbf{K}_L + \gamma \mathbf{I} & -\gamma \mathbf{I} \\ -\gamma \mathbf{I} & \mathbf{K}_R + \gamma \mathbf{I} \end{pmatrix}.$$

For $\mathbf{z} = (\mathbf{z}_L, \mathbf{z}_R)$,

$$\mathbf{z}^\top \mathbf{M}_W \mathbf{z} = \sum_{c=L,R} \left(\left\| \mathbf{W}_c^{1/2} \mathbf{A} \mathbf{z}_c \right\|^2 + \left\| \mathbf{\Lambda} \mathbf{z}_c \right\|^2 \right) + \gamma \left\| \mathbf{z}_L - \mathbf{z}_R \right\|^2.$$

Conclusion

$\mathbf{M}_W \succeq \lambda_{\min}^2 \mathbf{I} \succ 0$: the frozen quadratic has exactly one global minimizer.

Dissertation: Eq. 4.34, Theorem 4.6.3, Theorem 4.6.11.

Motion-induced events and the aperture problem

For a straight edge under steady illumination, let $\mathbf{n}$ be its unit normal, $\mathbf{t}_\perp$ a unit tangent, and s its normal speed:

$$L(\mathbf{x}, t) = \ell(\mathbf{n} \cdot \mathbf{x} - st), \qquad \nabla L = \ell' \mathbf{n}, \qquad L_t = -s\ell'.$$

Where $\ell' \neq 0$, brightness constancy gives

$$L_t + \nabla L \cdot \mathbf{v} = 0 \iff \mathbf{n} \cdot \mathbf{v} = s \iff \boxed{\mathbf{v} = s\mathbf{n} + \tau \mathbf{t}_\perp, \quad \tau \in \mathbb{R}.}$$

- Every such constant velocity gives the same local pattern and ideal events from the same reset reference.
- For $s = 0$, pixel values stay constant in the ideal edge interior; endpoints lie outside the observation window.

Takeaway. The edge determines normal speed; its tangential velocity remains free. Illumination changes can also trigger events under the sensor threshold rule.

Dissertation: Proposition 3.3.4 and following aperture remark; Eq. 2.1.

References (1 of 2)

[1] Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J. Davison, Jörg Conradt, Kostas Daniilidis, and Davide Scaramuzza. Event-based vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(1):154–180, 2022.

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